water and the other in freezing ice water, and then declares that the temperature is fine "on average."

I believe that a better way to think about longevity risk and uncertainty is via actuarial probability tables, such as Table 7.1.

TABLE 7.1 Probability of Survival at Age 65

To Age:	Female	Male	At Least One Member of a Male-Female Couple
70	93.9%	92.2%	99.5%
75	85.0%	81.3%	97.2%
80	72.3%	65.9%	90.6%
85	55.8%	45.5%	75.9%
90	34.8%	23.7%	50.3%
95	15.6%	7.7%	22.1%
100	5.0%	1.4%	6.3%

Using a "moderate" mortality assumption.

Data Source: RP2000 Mortality table; IFID Centre calculations.

For example, Table 7.1 shows that if you are a 65-year-old male, using a moderate mortality assumption, there is a greater than 45% chance that you will live to the age of 85. That would obviously require 20 years of retirement income, if you decide to retire exactly at the age of 65. Likewise, the same 65-year-old male has a 24% chance of living to the age of 90, which necessitates 25 years of income. For females the numbers are higher. A female who is 65 years of age has roughly a 35% chance of living to 90. Compare this number to the 24% probability for a male, and you can see the relative impact and magnitude of female longevity. In other words, if you have a large group of 65-year-old males, then slightly less than a quarter of them will live to the age of 90. Of course, we can't know in advance who will be included in that lucky quarter, so to be prudent they all plan for the possibility of 25 years of retirement income. The number increases significantly to 50%, if you consider the chances that at least one member of a male-female couple will survive to age 90.

Another way to think about longevity risk is by interpreting the risk in a more pessimistic manner. Indeed, according to the same